

AZ-400: Designing & Implementing DevOps Solutions

Complete Training Syllabus · Updated 2026

🕒 Duration: 10 Weeks

📜 Certification: DevOps Engineer Expert (AZ-400)

🔗 Prerequisite: AZ-104 or AZ-204

01

MODULE 01

DevOps Foundations & Process Design

- DevOps Culture, Practices & Principles
- Choosing the Right Project & Team Structure
- Azure Boards & Work Item Tracking
- GitHub Projects & Issue Management
- Communication, Collaboration & Documentation Strategy
- Migrating from Legacy Tools to Azure DevOps/GitHub

02

MODULE 02

Source Control Strategy

- Git Fundamentals & Enterprise-Scale Git
- Trunk-Based, GitFlow & GitHub Flow Strategies
- Branching Policies & Pull Request Workflows
- Managing Large Repositories & Git LFS
- Authentication & Authorization for Repos
- Recovering & Securing Repository Data

03

MODULE 03

Continuous Integration Pipelines

- Designing a CI Strategy (Azure Pipelines & GitHub Actions)
- YAML Pipelines, Templates & Reusable Components
- Self-Hosted vs. Microsoft-Hosted Agents/Runners
- Build Strategy for Containers
- Automated Testing in the Pipeline (Unit, Integration, Load)
- Pipeline Triggers, Variables & Conditions

04

MODULE 04

Release & Deployment Strategies

- Designing a Release Pipeline & Environments
- Deployment Patterns: Blue-Green, Canary & Ring-Based
- Feature Flags & Progressive Exposure
- Release Gates & Approval Workflows
- Rollback & Disaster Recovery Strategy
- Multi-Stage YAML Release Pipelines

05

MODULE 05

Dependency Management & Infrastructure as Code

- Package Management Strategy (NuGet, npm, Maven)
- Azure Artifacts & GitHub Packages
- Infrastructure as Code with ARM, Bicep & Terraform
- Configuration Management & Idempotency
- Provisioning Environments via Pipelines
- Policy-as-Code & Compliance Automation

Container Strategy

- Containerization Fundamentals for CI/CD
- Azure Container Registry & Image Management
- Building & Publishing Container Images in Pipelines
- Deploying to Azure Kubernetes Service (AKS)
- Container Build Optimization & Caching
- Managing Secrets in Containerized Deployments

Security & Compliance Plan

- Shift-Left Security & DevSecOps Principles
- GitHub Advanced Security (Code, Secret & Dependency Scanning)
- Microsoft Defender for Cloud DevOps Security
- Managing Secrets with Azure Key Vault
- Gating Releases on Security & Compliance Checks
- Governance, RBAC & Audit Strategy

Instrumentation, SRE & Continuous Feedback

- Implementing Application Insights & Azure Monitor
- Defining SLIs, SLOs & Error Budgets (SRE Strategy)
- Alerting on Deployment Failures & Regressions
- KQL Basics for Log Analytics
- Release Health Metrics: Lead Time, MTTR, Deployment Frequency
- Continuous Improvement & Feedback Loops

What You'll Walk Away With

End-to-end DevOps training covering source control, CI/CD pipelines, containers, Infrastructure as Code, security, and SRE-driven monitoring — built to get you certification-ready and job-ready.

[Live Mentorship](#)[Real CI/CD Labs](#)[Azure DevOps + GitHub](#)[LMS Access for Life](#)